

Notes: Kim, Pence, Stanton, Walden, and Wallace (ARFE, 2022)

Nonbanks and Mortgage Securitization

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1 Big picture

1.1 Question

Kim, Pence, Stanton, Walden, and Wallace ask how the post-crisis U.S. mortgage market works when mortgage lending is increasingly done by **nonbanks** and funded through **government-backed securitization**, especially Ginnie Mae pools.

The central question is not just why nonbanks grew. It is whether the new nonbank–Ginnie Mae mortgage system creates different screening, prepayment, default, liquidity, and capital risks than the older bank-centered securitization system studied in much of the literature.

The paper is partly a review article and partly an original analysis. It reviews securitization and nonbank mortgage lending, builds a simple model of bank versus nonbank mortgage origination, tests key mortgage-performance predictions, and uses regulatory data to study the balance sheets of large nonbank Ginnie Mae issuers.

1.2 Setting

The setting is the U.S. mortgage market after the Global Financial Crisis. The market is enormous: at year-end 2021, one-to-four-family residential mortgages were about 12.5 trillion dollars, and only about 3.2 trillion dollars were funded by banks. Most mortgage credit is therefore funded through capital markets and mortgage-backed securities.

The institutional structure changed sharply after the crisis. Private-label securitization collapsed and remained small. Lower-credit-quality mortgages became concentrated in Ginnie Mae pools, and nonbanks became dominant originators and servicers. The authors report that nonbanks originated about **60% of mortgages in 2020**, compared with **35% in 2013**, and about **90% of Ginnie Mae mortgages in 2020**, compared with **35% in 2013**.

1.3 Core challenge

The key challenge is that the literature that explains securitization problems before the crisis mostly studies **banks**, **private-label securities**, and **credit-risk retention**. The current system is different. It is built around nonbank mortgage companies, GSE and Ginnie Mae guarantees, mortgage servicing rights, short-term warehouse funding, and the liquidity obligations of servicers.

Interpretation. The paper’s economic point is that “skin in the game” is no longer mainly a retained credit tranche. For many nonbank issuers, the practical skin in the game is the mortgage servicing right, because servicing defaulted mortgages is costly and liquidity intensive.

1.4 Main contribution

- It reframes mortgage securitization around **nonbank** incentives rather than only bank incentives.

- It builds a simple model in which banks and nonbanks differ in capital regulation, speed of securitization, exposure to default, and incentives to collect soft information.
- It shows empirically that nonbank-originated mortgages prepay earlier than bank-originated mortgages, and Ginnie Mae nonbank loans also default earlier.
- It uses Mortgage Call Report regulatory filings, which are typically unavailable to researchers, to study the assets, liabilities, liquidity, and capital of the 100 largest nonbank Ginnie Mae forward-mortgage issuers.
- It highlights a policy trade-off: stricter capital rules may reduce systemic fragility but can also affect the liquidity of the market for mortgage servicing rights and the operational capacity of mortgage servicing.

1.5 Main result

The model predicts that nonbanks will lend more aggressively, screen less using costly soft information, have higher default and prepayment rates in their loan pools, and be more likely to exit or fail. The empirical evidence is consistent with several of these predictions.

- Nonbank loans are more likely to prepay within 24 months than bank loans, even after controlling for interest rates, FICO–LTV buckets interacted with origination month, and loan-amount deciles.
- The prepayment gap is larger for Ginnie Mae loans than for GSE loans.
- Ginnie Mae loans issued by nonbanks are also more likely to default early; the paper does not find a comparable significant default difference for GSE loans.
- Large nonbank Ginnie Mae issuers generally had stronger liquidity and capital by 2022:Q1 than before the pandemic, partly because refinancing profits increased cash.
- However, some large issuers look undercapitalized under risk-weighted measures that penalize mortgage servicing rights, and these issuers hold a large share of Ginnie Mae servicing unpaid principal balance.

1.6 Economic takeaway

Securitization lets nonbanks compete in mortgage origination because it gives them access to long-term capital-market funding. But it also creates a mortgage system in which thinly capitalized, mortgage-focused institutions originate and service many loans, especially loans to lower-credit-score borrowers. The central fragility is not only origination quality. It is also whether nonbank servicers have enough liquidity and capital to absorb delinquency, servicing, and MSR valuation shocks.

2 Data and measurement

2.1 Institutional objects

The paper repeatedly uses several mortgage-market objects. Understanding them is essential before reading the model and evidence.

- **Bank lender.** In the paper, banks include banks and credit unions. Banks are more regulated, have broader franchise value, have access to different funding sources, and may be more reluctant to take actions that create litigation, putback, or supervisory risk.
- **Nonbank lender.** Nonbanks are generally mortgage finance companies. Many are monoline mortgage firms. They rely heavily on securitization, warehouse lines, and servicing income; they do not have insured deposits or Federal Reserve liquidity access like banks.
- **GSE MBS.** Securities backed by loans that meet Fannie Mae or Freddie Mac standards. The GSEs guarantee timely principal and interest to investors, so investors bear prepayment risk but not ordinary mortgage credit risk.
- **Ginnie Mae MBS.** Securities issued by banks or nonbanks, with Ginnie Mae guaranteeing timely payment to investors. The underlying mortgages are often FHA or VA loans, which tend to serve lower-credit-score borrowers. The issuer/servicer faces important advancing and default-servicing obligations.
- **Private-label securities (PLS).** MBS without a GSE or Ginnie Mae guarantee. PLS were central before the Global Financial Crisis but became much smaller after the crisis.
- **Mortgage servicing rights (MSRs).** The present value of expected income from servicing a pool of mortgages. MSRs are assets on nonbank balance sheets, but they are volatile, illiquid in stress, and difficult to value.
- **Warehouse funding.** Short-term credit lines used to finance loans before securitization. Warehouse debt is central to the nonbank funding model and can create run-like fragility.

2.2 Loan-performance data

For the bank-versus-nonbank performance comparison, the authors use linked loan-level data from HMDA, ATTOM, and McDash. The linked data allow them to observe mortgage lender type, borrower and loan characteristics, contract terms, and loan performance.

- **HMDA** provides applicant income, race, ethnicity, loan amount, lender name, and property census tract.
- **ATTOM** provides property transaction and assessor information, lienholder names, borrower and lender names, loan performance indicators such as prepayment and default, exact location, loan purpose, origination date, and fixed-versus-floating contract type.
- **McDash** provides detailed contract and performance data, including interest rates, loan amount, loan-to-value ratio, zip code, and month-by-month loan performance.

The filtered performance sample focuses on first-lien, fixed-rate, owner-occupied, 30-year, single-family residential loans securitized by the GSEs or insured/guaranteed by FHA and VA, over 2005–2015. The authors exclude manufactured housing, investment properties, condos, multifamily properties, and loans with second liens at origination. The merge rate for candidate loans is about 73.99%.

2.3 Mortgage Call Report data

For nonbank liquidity and capital, the paper uses Mortgage Call Report filings. These regulatory filings are required under the SAFE Act for companies with state licenses or registrations through the Nationwide MultiState Licensing System. GSE seller/servicers and Ginnie Mae issuers file quarterly.

The authors study the **100 largest for-profit nonbank Ginnie Mae issuers of forward mortgages**. They exclude banks, reverse-mortgage specialists, state-affiliated mortgage revenue bond issuers, and captive finance arms of manufactured-home builders. They merge the MCR data with eMBS data on servicing portfolios and loan characteristics. The data cover 2015 through 2022:Q1.

The coverage is economically important. As of 2022:Q1, these firms held servicing rights on about **75% of loans in Ginnie Mae pools** and **39% of mortgages in Fannie Mae or Freddie Mac pools**.

2.4 Business-model measures

The nonbank business model is heterogeneous. Some firms primarily originate loans; some retain servicing; some buy MSRs from other firms; some subservice for others. The paper stresses that this heterogeneity matters because origination, MSR ownership, and servicing operations create different risks.

- **Retail originations** come directly from borrowers and are viewed as less risky because the lender has more direct information.
- **Broker originations** come through mortgage brokers.
- **Correspondent originations** are purchased from other firms that originated and underwrote the loans.
- **MSR aggregators** buy servicing rights from others; they can become systemically important buyers in the MSR market but can also hold large, volatile intangible assets.
- **Subservicing** means performing the operational servicing tasks for another MSR owner. It creates operational and compliance risk.

2.5 Liquidity and capital measures

The paper uses several measures to summarize nonbank financial strength.

- **Cash relative to quarterly expenses** is the main liquidity measure. It asks how many quarters of operating expenses a nonbank can cover with cash.

- **Ginnie Mae leverage ratio** treats equity relative to total assets and, as of June 2022, requires equity of at least 6% of total assets. This measure gives MSRs more credit as an asset.
- **Proposed Ginnie Mae risk-weighted capital ratio** requires a risk-weighted capital ratio above 10% and penalizes MSRs more heavily.
- **Bank Tier 1 risk-based capital ratio** is not designed for nonbanks, but it provides a benchmark for comparing banks and nonbanks. The paper uses the well-capitalized threshold of 8%.

Interpretation. The key distinction is whether MSRs are treated like high-quality capital support. The paper argues that this is questionable because MSRs may fall in value precisely when delinquencies rise and servicing liquidity needs increase.

3 Models and empirical specifications

3.1 Planner problem: hard information and soft information

The model starts with potential borrowers. A borrower is either vulnerable, denoted by ℓ , or resilient, denoted by h . A resilient borrower generates social value $A > 0$ if given a mortgage. A vulnerable borrower generates social cost $-B < 0$ if given a mortgage.

Each borrower has hard information $p \in (0, 1)$, interpreted like a credit score. Conditional on p , the borrower is resilient with probability p . The lender or planner can also acquire perfect soft information at cost $C > 0$.

$$p^* = \frac{B}{A + B}$$

Interpretation. Without soft information, the planner lends when the expected social benefit is positive, so lending starts above the hard-score threshold p^* . With soft information, the planner screens intermediate-score borrowers: some low-score resilient borrowers should receive loans, and some high-score vulnerable borrowers should be rejected.

The first-best allocation has three regions: no lending for very low scores; screening for intermediate scores; and automatic lending for sufficiently high scores.

$$\text{No loan if } p < C/A, \quad \text{screen if } C/A \leq p \leq 1 - C/B, \quad \text{lend if } p > 1 - C/B.$$

3.2 Mortgage technology and securitization

A mortgage gives the borrower 1 today. The borrower pays a perpetual mortgage payment stream at rate s . The discount rate is r , the lender's funding rate is q , and shocks arrive according to a Poisson process with intensity λ . The paper assumes $0 < r < q < s < r + \lambda$.

A vulnerable borrower defaults when a shock arrives; a resilient borrower continues paying. The lender funds the mortgage using equity e and debt $d = 1 - e$. If a vulnerable borrower defaults while the lender is levered, the lender may default as well.

A mortgage can be securitized after time T if it has not defaulted. The securitized mortgage is assumed to be fully insured by a GSE and is worth $P = s/r$.

$$V(p, e) = \left[pe^{-rT} + (1-p)e^{-(r+\lambda)T} \right] \left(\frac{s}{r} - d \right) + \left[p \frac{1 - e^{-rT}}{r} + (1-p) \frac{1 - e^{-(r+\lambda)T}}{r+\lambda} \right] (s - dq) - e.$$

Interpretation. The loan value has three pieces: the future sale value if the loan survives to securitization, the interim spread income before securitization, and the initial equity investment.

3.3 Banks

Banks are modeled as having long exposure to the mortgage before securitization, captured by the approximation $T = \infty$. This long exposure reflects slower balance-sheet removal, putback risk, regulatory scrutiny, and the fact that banks may hold or swap back agency securities.

$$V^B(p, e) = \frac{r + p\lambda}{r(r + \lambda)} [s - q(1 - e)] - e.$$

Interpretation. A bank faces a trade-off. More equity raises the stream of surplus dividends because debt funding is costly. Less equity reduces the equity lost if a vulnerable borrower defaults. High-score borrowers push banks toward high capital; low-score borrowers push banks toward minimum capital or no lending.

The model defines two bank thresholds:

$$p_1^* = \frac{r(r + \lambda - s)}{s\lambda}, \quad p_2^* = \frac{r(r + \lambda - q)}{q\lambda}.$$

For sufficiently high scores, the bank lends and chooses full equity funding. For lower scores, if it lends, it chooses the minimum allowed capital ratio. The minimum capital requirement therefore affects whether low-score borrowers get credit and whether the bank bears enough downside risk to screen.

3.4 Bank screening with soft information

A bank can pay C to learn the borrower's type. If it screens, it lends only when the borrower is resilient. The expected payoff from screening is:

$$V_C = p \left(\frac{s}{r} - 1 \right) - C.$$

Interpretation. Screening is attractive when the borrower is neither obviously bad nor obviously good. For very low scores, the chance of a resilient borrower is too small. For very high scores, screening is unnecessary. For intermediate scores, soft information helps separate resilient from vulnerable borrowers.

A key implication is that lowering capital requirements may expand lending to low-score borrowers but can also reduce screening. The bank may lend without acquiring soft information because the downside is shifted away from equity holders.

3.5 Nonbanks

Nonbanks differ from banks in two dimensions. First, they face lower minimum capital ratios, denoted $e_{NB} < e$. Second, they can sell or securitize mortgages faster. The time to sale is $S < T$, and the model focuses on the case where S is small.

$$V^{NB} \approx \left(\frac{s}{r} - 1\right) - \lambda(1-p)\frac{s}{r}S + (e-1)[q-r-\lambda(1-p)]S.$$

Interpretation. Because nonbanks sell loans quickly, they face little exposure to default before securitization. That makes costly soft-information acquisition less attractive. In the model, when S is small, nonbanks never screen and lend to all applicants.

The nonbank capital choice depends on the sign of $q - r - \lambda(1 - p)$. The relevant threshold is:

$$p_3^* = \frac{r + \lambda - q}{\lambda}.$$

For scores below p_3^* , the nonbank chooses the minimum capital ratio e_{NB} . For scores above p_3^* , it chooses full capital. Since $p_3^* = (q/r)p_2^* < p_2^*$, the nonbank is less selective and more aggressive over the intermediate-score region.

3.6 Model implications

1. Bank-originated mortgages default less often than nonbank mortgages, especially in distress.
2. Bank-originated mortgages prepay more slowly than nonbank mortgages.
3. Banks are less likely to go out of business than nonbanks.
4. Bank origination is more costly and labor intensive because banks are more likely to use soft-information screening.
5. Banks and nonbanks may choose similar capital for very low-score and very high-score borrowers, but nonbanks choose lower capital and lend more aggressively for intermediate-score borrowers.
6. Bank screening intensity may be nonmonotone in borrower score.

3.7 Empirical prepayment regression

The main empirical test of mortgage performance focuses on early prepayment. The regression is:

$$\text{Prepay24}_i = \alpha + \beta \text{Nonbank}_i + \gamma \text{Rate}_i + \mu_{m(i) \times g(i)} + \delta_{a(i)} + \varepsilon_i.$$

Interpretation. Prepay24_i equals one if loan i prepays within 24 months. The coefficient beta compares nonbank and bank loans within the same origination-month and GSE credit-pricing grid bucket and loan-amount decile, controlling for the loan rate.

The paper runs this regression separately for Ginnie Mae loans and GSE loans. A parallel specification is used for early default in the supplemental results.

3.8 Capital and liquidity analysis

The balance-sheet analysis is descriptive rather than a causal design. It asks whether the nonbank institutions that dominate Ginnie Mae servicing appear to have enough cash and capital under plausible regulatory standards.

$$\text{Liquidity} = \frac{\text{cash and cash equivalents}}{\text{quarterly expenses}}.$$

$$\text{Ginnie Mae leverage ratio} = \frac{\text{equity}}{\text{total assets}} \quad \text{with required minimum 6\%}.$$

Risk-based capital ratios penalize MSRs by subtracting or heavily risk-weighting them.

Interpretation. The difference between leverage and risk-weighted measures is central. If MSRs are treated as normal assets, most issuers look healthy. If MSRs are treated as risky and illiquid, some large issuers look undercapitalized.

4 Identification and interpretation

4.1 What is identified in the loan-performance exercise?

The loan-performance analysis is not a randomized experiment. It is best read as a disciplined comparison of bank and nonbank loans with rich observable controls, designed to test the model's predictions.

The specification controls for several obvious sources of selection: origination date, interest rate, credit-score and LTV bucket, GSE pricing grid, and loan amount. The remaining coefficient on Nonbank captures differences not explained by those observables.

Interpretation. The evidence says that even among observably similar loans, nonbank loans are more likely to prepay early. This supports the model's idea that nonbanks originate pools with different risk characteristics or different borrower/lender incentives than banks.

4.2 Why the prepayment result is informative

Prepayment is especially important in agency mortgage markets because default risk is largely insured by the GSEs or the government, while prepayment risk is borne by MBS investors. Therefore, prepayment differences can affect investor returns even when credit losses are guaranteed.

The paper argues that the prepayment gap is not simply due to nonbank borrowers making more rational interest-rate refinancing decisions. The authors show that the pattern appears across issue years, including years when interest rates rose and early prepayment was unlikely to be driven primarily by rate incentives.

4.3 What is identified in the capital analysis?

The capital analysis identifies the financial condition of the large nonbank Ginnie Mae issuers in regulatory data. It does not estimate a causal effect of capital on servicing performance. Its role

is diagnostic: it shows whether systemically important servicers appear robust under alternative capital definitions.

The key empirical contrast is between the current Ginnie Mae leverage ratio and risk-weighted capital measures. The current leverage ratio gives MSRs substantial asset value; risk-weighted measures treat MSRs as fragile.

Interpretation. This is a policy-relevant measurement issue. If MSRs retain value in stress, nonbanks look safer. If MSRs fall in value exactly when defaults and servicing advances rise, the same balance sheets look more fragile.

4.4 Main limitations

- The model is highly stylized. It assumes exogenous mortgage rates, perfect soft information, and simplified default and securitization timing.
- The empirical prepayment and default comparisons can still be affected by unobserved borrower selection or lender-specific behavior not captured by the controls.
- Mortgage Call Report data provide rare coverage of private nonbanks, but regulatory reporting categories are imperfect proxies for economic exposures.
- There is no universally accepted capital framework for nonbank mortgage servicers, so the paper compares several plausible metrics rather than claiming one definitive rule.
- MSR values are model-based and can be difficult to measure, especially because MSRs trade infrequently and may become illiquid during stress.

4.5 How to read the paper’s policy argument

The paper does not simply say “raise capital requirements.” It says that the optimal policy is a trade-off. Nonbanks need enough liquidity and capital to service delinquent loans and avoid disrupting mortgage credit. But the firms most affected by stricter capital rules are often the same firms that provide liquidity in the MSR market and operational servicing capacity.

The policy problem is therefore to make nonbank mortgage intermediation safer without accidentally shrinking mortgage servicing capacity or making the MSR market less liquid during normal times.

5 Tables and figures

5.1 Figure 1: cumulative prepayment by bank versus nonbank issuer

Read: The two panels compare bank and nonbank cumulative prepayment for Ginnie Mae loans and GSE loans. In both panels, nonbank loans prepay faster. The gap is especially visible for Ginnie Mae loans.

Economic message: This is the cleanest visual evidence for the model’s prediction that nonbanks originate loans with higher prepayment risk. Because prepayment risk matters to MBS investors, the nonbank-bank difference is economically important even when credit risk is guaranteed.

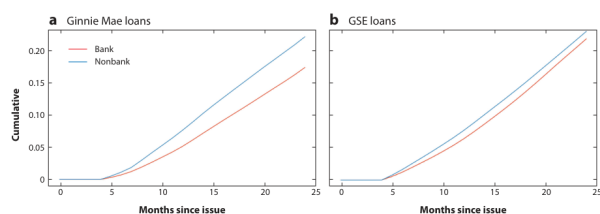


Figure 1 Cumulative prepayment in the first 24 months after issue for loans issued by banks versus nonbanks. (a) Ginnie Mae loans. (b) GSE

Table 1 Prepayment

Variables	Ginnie Mae loans prepay24	GSE loans prepay24
Nonbank	0.0161*** (0.000442)	0.0104*** (0.000331)
Loan interest rate	15.49*** (0.0686)	12.06*** (0.0502)
Observations	2,262,186	4,606,484
R-squared	0.125	0.105
FICO/LTV bucket $\times$ year/month FE	Y	Y
Amount decile FE	Y	Y

The dependent variable equals 1 if the loan prepaid within 24 months of issue, and 0 otherwise. The independent variable *nonbank* equals 1 if the lender is a nonbank institution, and 0 otherwise. FE are included for year/month $\times$ GSE-grid bucket and loan-amount decile. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels. Abbreviations: FE, fixed effect; GSE, government-sponsored enterprise; LTV, loan-to-value ratio.

5.2 Table 1: prepayment regression

Read: The nonbank coefficient is 0.0161 for Ginnie Mae loans and 0.0104 for GSE loans. Both estimates are statistically significant. The regressions include origination-year/month by credit-score/LTV bucket fixed effects and loan-amount decile fixed effects.

Economic message: Even after matching loans on important observable pricing and timing dimensions, nonbank loans prepay earlier. The coefficient is larger for Ginnie Mae loans, consistent with greater risk concentration in the nonbank-Ginnie Mae segment.

5.3 Figure 2: nonbank asset composition

Read: The figure decomposes assets into cash, mortgage servicing rights, loans held for sale, and loans eligible for repurchase. The balance sheet is dominated by mortgage-related assets rather than diversified assets.

Economic message: The key issue is asset quality under stress. Loans held for sale are pledged to warehouse lenders, loans eligible for repurchase are accounting assets tied to delinquent loans, and MSRs are volatile and potentially illiquid. Cash is the cleanest liquidity buffer but is a small share of assets.

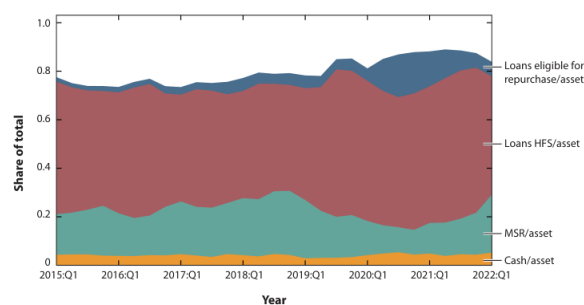


Figure 2 Nonbank asset composition. Abbreviations: loans HFS, loans held for sale; MSR, mortgage servicing right. From staff calculations based on data from the Conference of State Bank Supervisors, Nationwide MultiState Licensing System & Registry (<https://nationwidelicensingsystem.org/slr/common/mcr/Pages/default.aspx>).

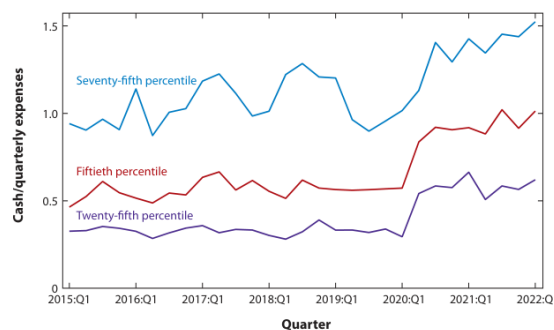


Figure 3 Percentiles of cash relative to quarterly expenses. From authors' calculations based on data from the Conference of State Bank Supervisors, Nationwide MultiState Licensing System & Registry (<https://nationwidelicensingsystem.org/slr/common/mcr/Pages/default.aspx>).

5.4 Figure 3: cash relative to expenses

Read: Cash relative to quarterly expenses is low before the pandemic, rises substantially in 2020 and 2021, and remains higher by 2022:Q1. The median issuer has about one quarter of expenses in

cash by 2022:Q1.

Economic message: The pandemic refinancing wave temporarily strengthened liquidity. But the pre-pandemic pattern shows that many servicers operated with thin cash buffers, and the lower tail still has limited cash relative to expenses.

5.5 Figure 4: median capital ratios under alternative rules

Read: The median of all three ratios remains above relevant thresholds. The medians rise after the pandemic, likely due to refinancing profits and higher cash generation.

Economic message: On average, large nonbank Ginnie Mae issuers do not look severely undercapitalized. The concern is not the median firm; it is the lower tail and the firms with large servicing books.

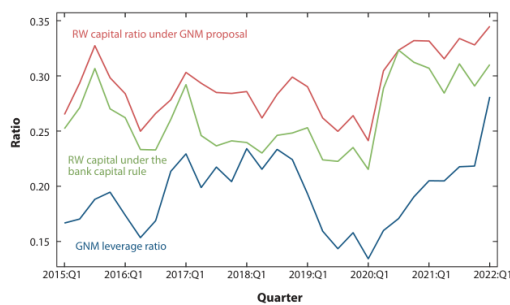


Figure 4
Median capital ratio under different rules. Abbreviations: GNM, Ginnie Mae; RW, risk weighted. From authors' calculations based on data from the Conference of State Bank Supervisors, Nationwide MultiState Licensing System & Registry (<https://nationwidelicensingsystem.org/slr/common/mcr/Pages/default.aspx>).

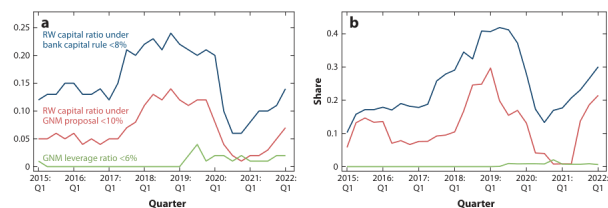


Figure 5
Share of nonbanks with binding capital rules. (a) Equal weighted. (b) Weighted by Ginnie Mae servicing unpaid principal balance. Abbreviations: GNM, Ginnie Mae; RW, risk weighted. From authors' calculations based on data from the Conference of State Bank Supervisors, Nationwide MultiState Licensing System & Registry (<https://nationwidelicensingsystem.org/slr/common/mcr/Pages/default.aspx>).

5.6 Figure 5: share of nonbanks with binding capital rules

Read: Almost no issuers fail the current Ginnie Mae leverage ratio. But a much larger share fail risk-weighted measures, especially in 2018–2019 and especially when weighted by servicing UPB.

Economic message: Risk-weighted capital measures reveal concentrated fragility. The firms below thresholds are not necessarily small marginal players; some control large servicing portfolios, so their weakness could matter for the mortgage system.

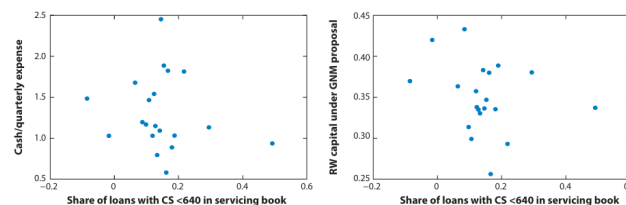


Figure 6
Liquidity and capital by share of servicing portfolio with low CS. (a) Cash/quarterly expense. (b) RW capital under GNM proposal. Abbreviations: CS, credit score; GNM, Ginnie Mae; RW, risk weighted. Data are for 2022:Q1. From authors' calculations based on data from the Conference of State Bank Supervisors, Nationwide MultiState Licensing System & Registry (<https://nationwidelicensingsystem.org/slr/common/mcr/Pages/default.aspx>) and eMBS, Inc.

5.7 Figure 6: liquidity and capital versus low-credit-score exposure

Read: The scatterplots show no clear relationship between the share of low-credit-score loans in the servicing book and either cash relative to expenses or risk-weighted capital.

Economic message: This is important because low-credit-score loans create greater default-servicing and advance risks. The absence of a clear relation suggests that neither regulation nor market convention makes servicers provision more when their portfolio is riskier.

6 Short summary

6.1 One-paragraph summary

Kim et al. study the post-crisis mortgage system in which nonbank lenders dominate origination and servicing, especially in Ginnie Mae pools. They argue that the traditional securitization literature is incomplete for this setting because nonbank incentives are shaped by fast securitization, weaker prudential oversight, lower franchise value, short-term funding, MSRs, and servicing obligations. Their model predicts that nonbanks lend more aggressively, screen less, and originate loans with higher default and prepayment risk. Their loan-level evidence shows that nonbank loans prepay earlier than bank loans and that Ginnie Mae nonbank loans also default earlier. Their regulatory balance-sheet evidence shows that large Ginnie Mae nonbank issuers were generally more liquid and better capitalized by 2022:Q1 than before the pandemic, but that some large issuers look weak under risk-weighted capital measures, especially because MSRs are large and fragile assets.

6.2 Main lessons

- The U.S. mortgage market is now a nonbank-securitization market, not mainly a bank balance-sheet market.
- Ginnie Mae is central because lower-credit-score mortgages are increasingly funded through Ginnie Mae pools after the collapse of private-label securitization.
- Nonbanks have weaker incentives to acquire soft information because they sell loans quickly and bear less default exposure before securitization.
- Prepayment risk is a key investor risk in agency MBS markets, and nonbank-originated loans prepay faster.
- The key nonbank balance-sheet asset is the MSR, but MSRs are risky because they are volatile, illiquid, and vulnerable in stress.
- Current leverage-style capital measures can make nonbanks look safer than risk-weighted measures that penalize MSRs.
- The lowest-capital large issuers are important for MSR-market liquidity and servicing operations, which complicates policy design.

6.3 What to remember

- **Conceptual innovation:** bring nonbanks into the theory of mortgage securitization.
- **Empirical fact:** nonbank loans prepay faster than bank loans; Ginnie Mae nonbank loans also default more.
- **Measurement innovation:** use Mortgage Call Reports to observe private nonbank balance sheets.
- **Policy insight:** nonbank regulation must account for servicing advances and MSR fragility, not just origination volume.

7 Conclusion

The paper's conclusion is that the post-crisis mortgage system has moved risk into a new institutional form. The old private-label securitization market is no longer the main issue. The central issue is a nonbank-dominated agency and Ginnie Mae system in which origination, servicing, liquidity, capital, and mortgage servicing rights interact.

Nonbanks make mortgage credit more available because securitization allows them to compete with banks. But the same structure can also create fragility. Nonbanks rely on short-term funding, hold mortgage-concentrated assets, and must perform servicing obligations during stress. Their key asset, the MSR, can lose value or become illiquid precisely when they need liquidity most.

The empirical results support the model's basic prediction: nonbank loans exhibit higher early prepayment risk and, in the Ginnie Mae segment, higher early default risk. The balance-sheet evidence shows that the median issuer may look adequate, but that some large issuers with important servicing roles appear weak under more conservative capital measures.

The final takeaway is a regulatory trade-off. Policymakers need nonbank servicers to have enough capital and liquidity to keep the mortgage system functioning in stress. But they must also preserve servicing capacity and MSR-market liquidity. The paper therefore frames nonbank mortgage regulation as a system-design problem, not simply a question of whether nonbanks should be bigger or smaller.